

B105 Applied Statistical Modeling
Bike Demand Analysis
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Table of contents:

- 1. Introduction & Business Question**
- 2. Methods**
- 3. Results & Discussion**
- 4. Recommendations & Conclusion**
- 5. Appendix**
- 6. References**

1. Introduction & Business Problem:

A Seoul bike share company struggles to assess the demand based on the season, the day, the hour of the day, and the weather. Furthermore, this matters because if the supply is less than the demand, this would result in a financial loss for the company. If the supply is higher than the demand, it results in higher operational costs and financial loss.

The objective of this project is to understand whether the season, the day, the hour of the day, and the weather actually have an impact on the bike demand, and to answer the following business question:

Do the following features: season, the day, the hour of the day, and weather-related features have a significant effect on bike demand?

2. Methods:

GitHub Repository link:

https://github.com/Khaled2027/Seoul_Bike_Demand_Analysis

Official Data Source:

<https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand>

The dataset for this project has been collected from the UCI Machine Learning Repository. The data is stored in one CSV file. It contains 8760 rows and 14 columns. Hence, sampling is not needed.

These are the variables that are given in the dataset:

1. Date: originally stored as string/character column stored in day/month/year format.
2. Rented Bike Count: continuous variable.
3. Hour: Discrete variable.
4. Temperature(°C): continuous variable.
5. Humidity(%): continuous variable.
6. Wind speed (m/s): continuous variable.
7. Visibility (10m): continuous variable.
8. Dew point temperature(°C): continuous variable.
9. Solar Radiation (MJ/m²): continuous variable.
10. Rainfall(mm): continuous variable.
11. Snowfall (cm): continuous variable.
12. Seasons: categorical variable.
13. Holiday: categorical variable.
14. Functioning Day: categorical variable.

```

# -----Data Preprocessing -----

# Keeps rows where 'Functioning Day' column is equal to 'Yes'
df <- df %>%
  filter('Functioning Day'=='Yes')
dim(df)
# Drops the 'Dew point temperature(°C)' column and 'Functioning Day' column
# from the dataframe
df <- df %>%
  select(!c('Dew point temperature(°C)', 'Functioning Day'))

# Switches the column data format into date format
df <-df %>%
  mutate(Date = as.Date(Date,format='%d/%m/%Y'))

# Returns a new column where days are numeric (Monday = 1, Sunday = 7)
df <-df %>%
  mutate(dayofweek = wday(Date,label = TRUE))

# Creates a new column called 'is_weekday' that contain two values: 1 (True) and 0 (False)
df$is_weekday <- ifelse(df$dayofweek=='Fri'|df$dayofweek=='Mon'|df$dayofweek=='Tue'|df$dayofweek=='Wed'|df$dayofweek=='Thu',1,0)

# Creates a new column called 'is_holiday' for encoding and it contains two values:
# 1 (True) and 0 (False)
df$is_holiday <-ifelse(df$Holiday=='Holiday',1,0)

df$is_winter <- ifelse(df$Seasons=='Winter',1,0)
df$is_spring <- ifelse(df$Seasons=='Spring',1,0)
df$is_summer <- ifelse(df$Seasons=='Summer',1,0)

df$is_snowfall <-ifelse(df$`Snowfall (cm)`==0,"No snowfall","Snowfall")

# Converts the datatype of 'Rented Bike Count' column into numeric type
df <-df %>%
  mutate('Rented Bike Count' = as.numeric('Rented Bike Count'))
# Drops the Holiday, and dayofweek columns
df <- df %>%
  select(!c(Holiday,dayofweek))

173 #----- Modeling -----
174
175 df_linear_regression <- df %>%
176   select(!c(Date,Seasons,is_snowfall))
177 # Multivariable Linear Regression Model

```

The following columns were feature engineered:

is_weekday (binary categorical integer column): firstly, I took the date column and feature engineered Weekday column takes the date format day/month/year and assigns a label for the day of the week. Moreover, I used that column to feature engineer **is_weekday**: it contains two values: 1 (the day is on a weekday) and 0 (the day is on a weekend).

Is_snowfall (categorical integer column): it contains two values: 1 (snowfall has happened and is greater than or equal to 1) and 0 (snowfall equal to 0).

Is_winter (categorical integer column): it contains two values: 1 (winter is the season) and 0 (it is a different season).

Is_spring (categorical integer column): it contains two values: 1 (spring is the season) and 0 (it is a different season).

Is_summer (categorical integer column): it contains two values: 1 (summer is the season) and 0 (it is a different season). Additionally, when all three columns are equal to zero, this means the season is autumn.

Is_holiday (categorical integer column): it contains two values: 1 (holiday) and 0 (not holiday), which is derived from the Holiday column.

The column Functioning Day contains 2 unique values: 'Yes' and 'No'. I have filtered by 'Yes' since the project mainly focuses on days where there were bikes operating.

The column Rented bike count was a character type and it was later converted to a numerical column.

The following columns were dropped:

- Dew point temperature(°C): because this is derived from the temperature and humidity, also keeping it would mean having redundant information.
- Functioning Day: dropped after filtering, since all of its values were equal to Yes.
- Date: dropped after the data visualization, since the column is not an integer column that can be used for linear regression.

The following statistical tests will be implemented:

- **ANOVA:** Is there a difference in the average bike demand across seasons? If so, what specific seasons differ?
- **Two-sample t-test** (snowfall vs no snowfall, within Winter): is the average bike demand across snowing hours and non-snowing days the same or does it differ?
- **Two-sample t-test** (holiday vs not holiday): is the average bike demand across holiday and non-holiday days the same or does it differ?
- **Two-sample t-test** (weekday vs weekends): is the average hourly bike demand across weekdays and weekends the same or does it differ?
- **Multivariable Linear Regression:** Does the season, weather, or hour of the day significantly predict Rented bike demand? If so, what is each factor's effect?

3. Results & Discussion

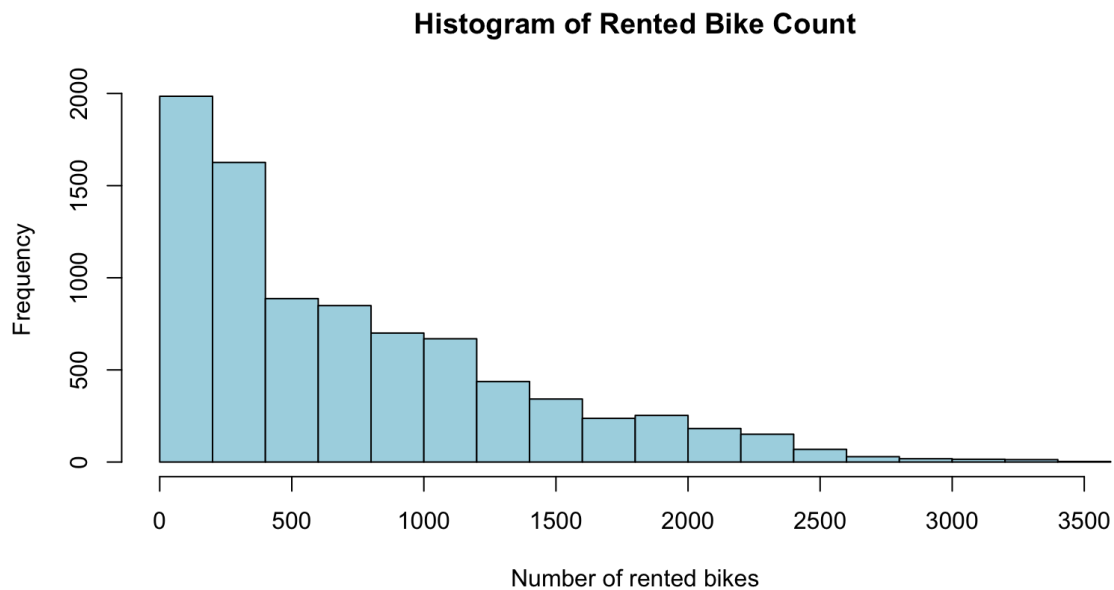
Throughout my data exploration journey, I used the describe function from the psych library only on numerical columns.

The results:

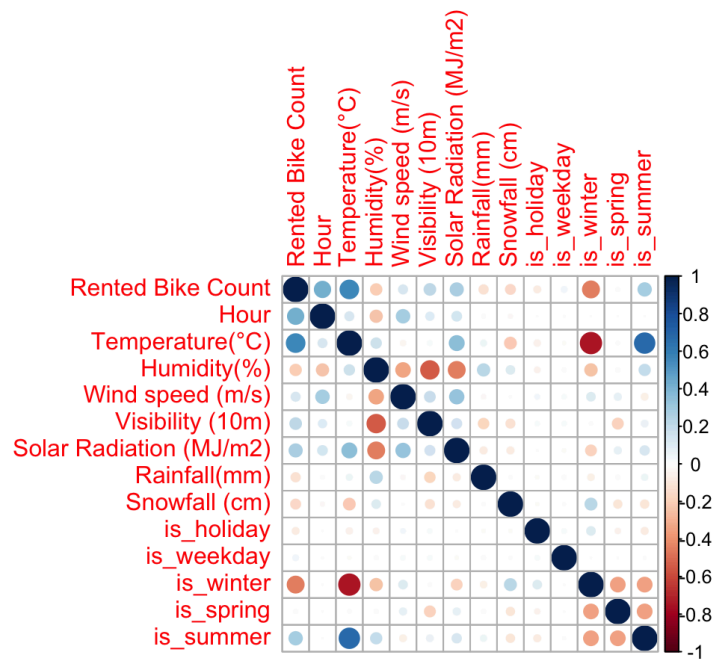
	vars	n	mean	sd	median	trimmed	mad	min
Rented Bike Count	1	8465	729.16	642.35	542.00	638.27	566.35	2.0
Hour	2	8465	11.51	6.92	12.00	11.51	8.90	0.0
Temperature(°C)	3	8465	12.77	12.10	13.50	13.05	14.38	-17.8
Humidity(%)	4	8465	58.15	20.48	57.00	57.92	23.72	0.0
Wind speed (m/s)	5	8465	1.73	1.03	1.50	1.63	1.04	0.0
Visibility (10m)	6	8465	1433.87	609.05	1690.00	1506.10	459.61	27.0
Solar Radiation (MJ/m2)	7	8465	0.57	0.87	0.01	0.38	0.01	0.0
Rainfall(mm)	8	8465	0.15	1.13	0.00	0.00	0.00	0.0
Snowfall (cm)	9	8465	0.08	0.44	0.00	0.00	0.00	0.0
is_holiday	10	8465	0.05	0.21	0.00	0.00	0.00	0.0
is_weekday	11	8465	0.71	0.45	1.00	0.76	0.00	0.0
is_winter	12	8465	0.26	0.44	0.00	0.19	0.00	0.0
is_spring	13	8465	0.26	0.44	0.00	0.19	0.00	0.0
is_summer	14	8465	0.26	0.44	0.00	0.20	0.00	0.0

	max	range	skew	kurtosis	se
Rented Bike Count	3556.00	3554.00	1.14	0.82	6.98
Hour	23.00	23.00	0.00	-1.20	0.08
Temperature(°C)	39.40	57.20	-0.17	-0.89	0.13
Humidity(%)	98.00	98.00	0.07	-0.81	0.22
Wind speed (m/s)	7.40	7.40	0.89	0.75	0.01
Visibility (10m)	2000.00	1973.00	-0.70	-0.97	6.62
Solar Radiation (MJ/m2)	3.52	3.52	1.51	1.14	0.01
Rainfall(mm)	35.00	35.00	14.61	289.72	0.01
Snowfall (cm)	8.80	8.80	8.29	90.47	0.00
is_holiday	1.00	1.00	4.22	15.79	0.00
is_weekday	1.00	1.00	-0.93	-1.13	0.00
is_winter	1.00	1.00	1.12	-0.74	0.00
is_spring	1.00	1.00	1.12	-0.74	0.00
is_summer	1.00	1.00	1.09	-0.81	0.00

The number of rows is 8465. Furthermore, for the Rented Bike Count column, the mean is at approx 729 rented bikes per hour, standard deviation at 642, median is at 542 bikes per hour, the minimum is at 2 bikes per hour, and the skewness value is at 1.14 which significantly implies the distribution is positively skewed. Kurtosis is 0.82 indicating heavy tails; hours with high demand that is not common.



This histogram shows that most bike demand is within the range of 0-1500, the most frequent range is from 0 to 250, followed by 250 to 500, and as the range increases the frequency decreases. There are also hours where the hourly bike demand is at 3000; this implies that there are hours where the demand is unusual.

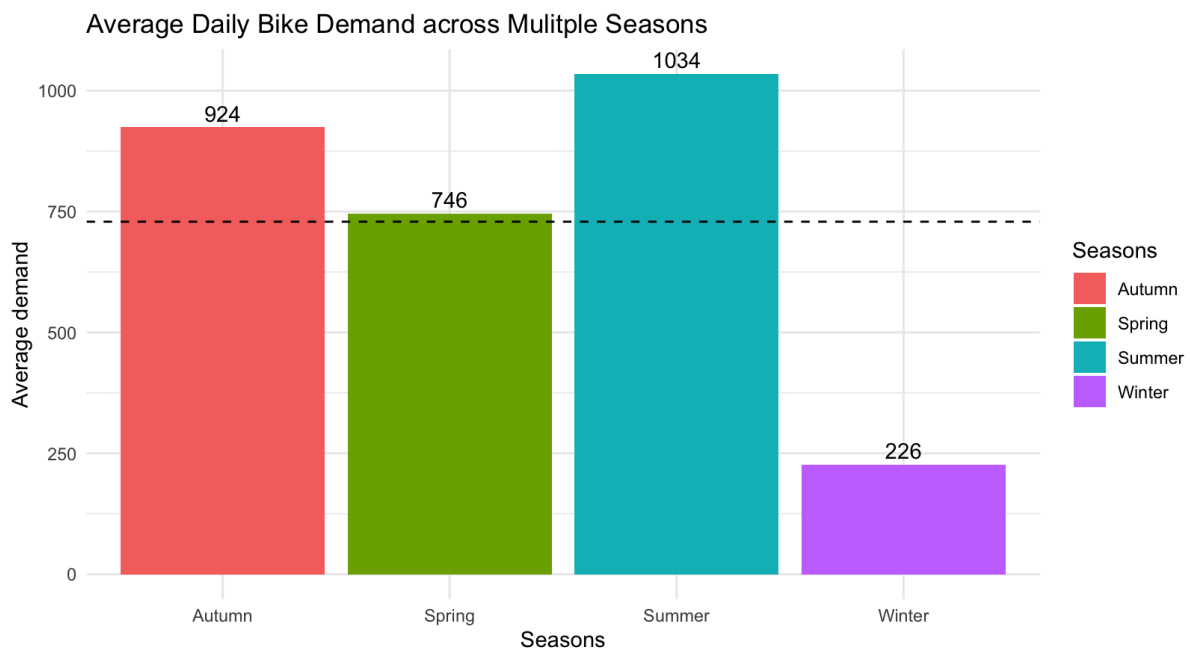


The figure above is a correlation matrix that shows us the strength of the relationship between 2 variables. The variable that has the strongest

relation with Rented Bike Count is Temperature at approx 0.4-0.6; the variable with the second strongest correlation is Hour at approx 0.2-0.4; visibility, wind speed, is_summer, is_weekday, and solar radiation all have values at approx 0.1-0.3. However, variables such as Humidity, Rainfall, Snowfall, and is_winter have a negative correlation in the approx range of -0.4 to -0.01.

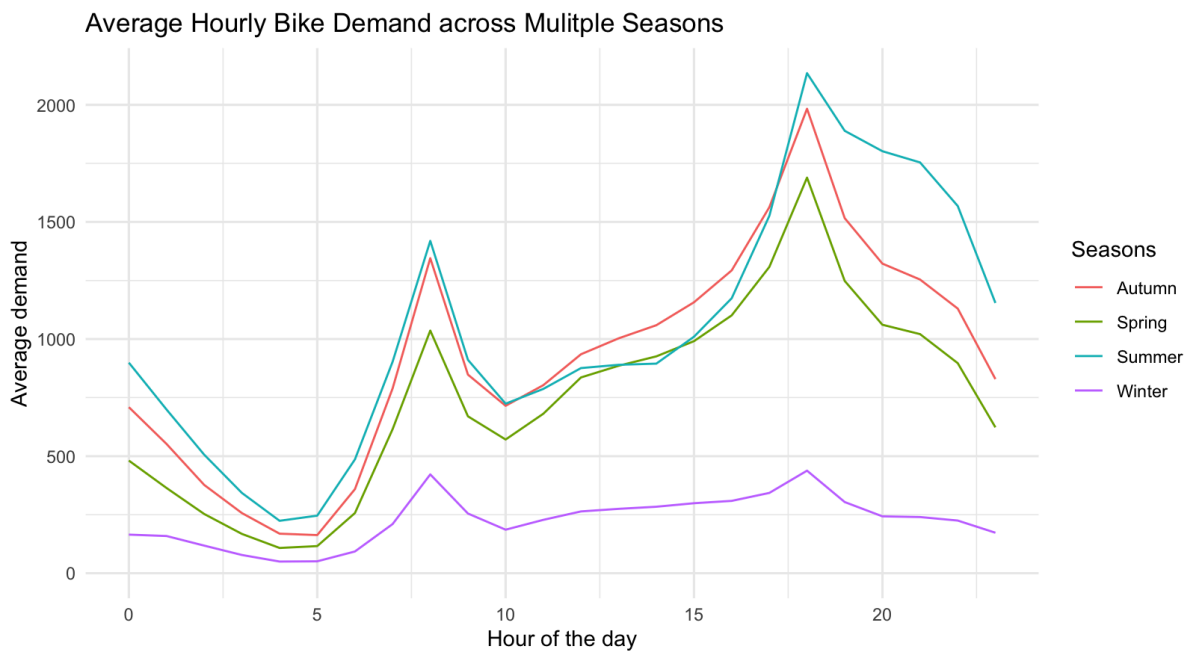
The following visuals below are presented for the Seoul Bike demand stakeholders:

What is the average bike demand across each season?



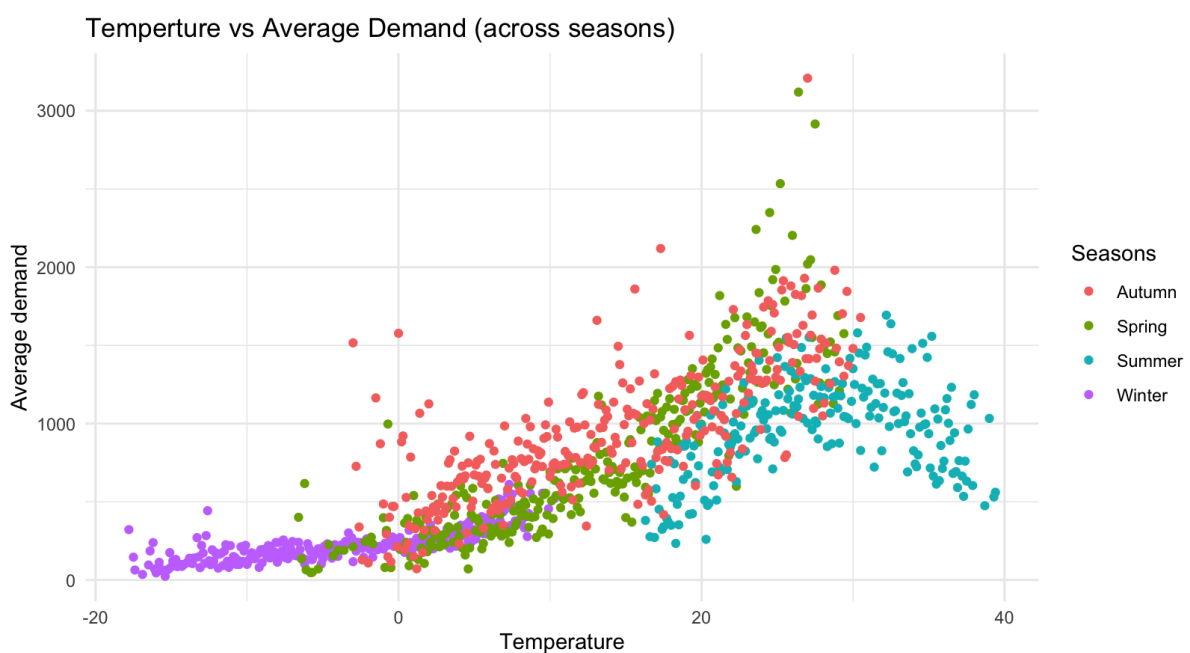
The figure above shows the average bike demand by season (where x represents the season and y represents the average demand). The season with the highest average bike demand is summer at 1034, way above the average of 729; autumn is in second place with 924, above average; in third place we have spring at 746, which is slightly above average; lastly, we have winter at 226, significantly below the average demand.

How does the hourly pattern of bike demand differ across each season?



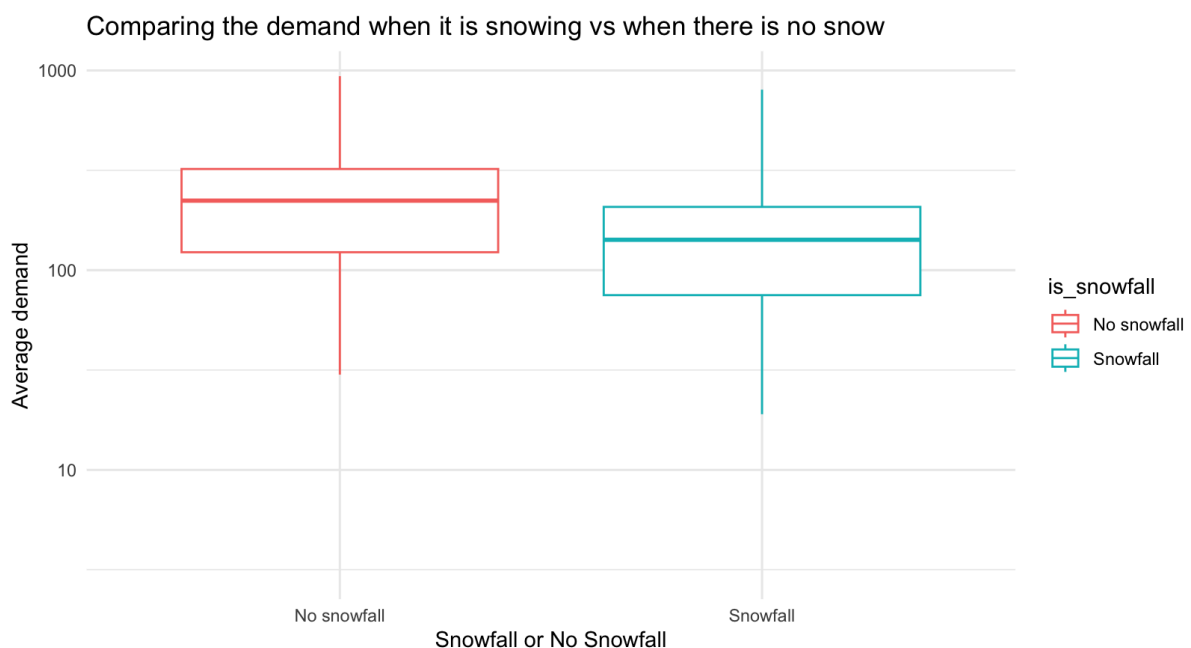
In this line chart above, x represents the hour of the day and y is the average demand. The seasons with the highest demand in most hours are summer, followed by autumn, spring, and lastly, winter. The chart illustrates an increased gap between average hourly bike demand in winter after hour 8.

Does average hourly temperature influence average bike demand, and does this relationship differ across seasons?



This scatterplot illustrates x as the average hourly temperature, and y as the average bike demand. This chart illustrates winter points: when the temperature increases for winter points, the average demand increases; however, for autumn, some points have higher demand than others with higher temperature, the same applies to spring, and in summary we can see that demand increases at the start as the temperature increases; however, when it reaches approximately 35, it starts to decrease. The relationship is mostly directly proportional: as temperature increases, bike demand tends to increase.

How does the bike demand differ between snowing and non-snowing hours within the Winter season?



The boxplot above illustrates: the first plot is no snowfall and the second is snowfall; the median is higher in non-snow hours at approx 250-260 and in snowing hours it is at approx 140-200; the maximum is higher for non-snowing hours at approx 800-900, and in snowing hours at 800-900; the minimum is higher for non-snowing hours at 50, while snowing is at approx 30; the interquartile range is wider for non-snowing hours at approx 120-300, and for snowing hours it is at approx 80-160.

- **Hypothesis test 1: ANOVA**

- H0 (null hypothesis): All seasons' 'rented bike count' means are equal.
- H1 (alternative hypothesis): At least one season's mean 'rented bike count' differs from the others.

Results:

```

              Df    Sum Sq   Mean Sq F value Pr(>F)
df$Seasons     3 8.274e+08 275791873   875.6 <2e-16 ***
Residuals  8461 2.665e+09   314974

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The results above illustrate that the F-value is at 875.6, which shows that the seasons significantly differ, and the p-value is less than 0.05. Hence, the null hypothesis is rejected.

Since ANOVA showed us that the groups significantly differ, Tukey's HSD was used to see the significance between each pair of seasons.

Results:

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = df$`Rented Bike Count` ~ df$Seasons)
```

```
$`df$Seasons`
```

	diff	lwr	upr	p adj
Spring-Autumn	-177.8563	-222.98313	-132.7295	0
Summer-Autumn	109.9629	65.06858	154.8572	0
Winter-Autumn	-698.5693	-743.69609	-653.4425	0
Summer-Spring	287.8192	244.17684	331.4616	0
Winter-Spring	-520.7130	-564.59447	-476.8315	0
Winter-Summer	-808.5322	-852.17453	-764.8898	0

Based on the statistical evaluation above, all six pairwise combinations show a p adj value of 0, indicating that every season differs significantly

from every other season in average bike demand. Winter shows the lowest demand overall, differing most dramatically from summer with a difference of -808.5322, followed by Autumn with a difference of -698.5693, and Spring by a difference of -520.7130. Summer constantly shows the highest demand, differing significantly from other seasons, which includes a 287.8192 difference from Spring and a 109.9629 difference from Autumn. Moreover, Autumn differed significantly from Spring with a difference of 177.8563.

These results confirm the statistical significance that was shown in the earlier visual (number 3). Winter represents the season with the lowest demand, Autumn has a significantly higher demand than Spring, and Summer represents the season with the highest demand.

Hypothesis Test 2: Welch Two Sample t-test

- H0 (null hypothesis): Within the winter season, the mean of 'rented bike count' during snowfall is equal to the mean of 'rented bike count' during No snowfall.
- H1 (alternative hypothesis): Within the Winter season, the mean of 'rented bike count' during snowfall is not equal to the mean of 'rented bike count' during No snowfall.

Results:

```
Welch Two Sample t-test

data: Rented Bike Count by is_snowfall
t = 12.587, df = 778.19, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No snowfall and group
Snowfall is not equal to 0
95 percent confidence interval:
 70.36614 96.36835
sample estimates:
mean in group No snowfall    mean in group Snowfall
      240.6708              157.3036
```

The statistical evaluation above illustrates that means significantly differ and that No snowfall has higher bike demand. As well, the p-value is less than 0.05. Therefore, the null hypothesis is rejected.

Hypothesis Test 3: Welch Two Sample t-test

- H0 (null hypothesis): The mean of rented bike count during holiday is equal to the mean of 'rented bike count' during not holiday.
- H1 (alternative hypothesis): The mean of rented bike count during holiday is not equal to the mean of 'rented bike count' during not holiday.

Results:

Welch Two Sample t-test

```
data: Rented Bike Count by is_holiday
t = 7.1703, df = 460.46, p-value = 3.002e-12
alternative hypothesis: true difference in means between group 0 and group 1 is not
equal to 0
95 percent confidence interval:
 152.5410 267.7201
sample estimates:
mean in group 0 mean in group 1
   739.2850      529.1544
```

The results illustrate that the mean for rented bike count differs significantly between holiday and not holiday, with no holiday showing the higher mean (739.2850) and holiday (529.1544); the p-value is significantly less than 0.05. Therefore, the null hypothesis is rejected. This indicates that the assumption that bike demand is higher on holidays vs non-holidays is false.

Hypothesis Test 4: Welch Two Sample t-test

- H0 (null hypothesis): The mean of rented bike count during weekdays is equal to the mean of 'rented bike count' during weekends.
- H1 (alternative hypothesis): The mean of rented bike count during weekdays is not equal to the mean of rented bike count during weekends.

Results:

Welch Two Sample t-test

```
data: Rented Bike Count by is_weekday
t = -4.4203, df = 4876.6, p-value = 1.007e-05
alternative hypothesis: true difference in means between group 0 and group 1 is not
equal to 0
95 percent confidence interval:
-94.88752 -36.57985
sample estimates:
mean in group 0 mean in group 1
682.3785      748.1122
```

The results illustrate that the mean for rented bike count differs significantly between weekdays and weekends, with weekends showing the higher mean (748.1122) and weekdays (682.3785); the p-value is significantly less than 0.05. Therefore, the null hypothesis is rejected.

Hypothesis Test 5: Multivariable Linear Regression

- H0 (null hypothesis): None of the independent variables have a significant effect on the rented bike count.
- H1 (alternative hypothesis): At least one of the independent variables has a significant effect on the rented bike count.

Results for VIF (Variance Inflation Factor):

Hour	Temperature(°C)	Humidity(%)
1.205115	5.109573	2.630479
Wind speed (m/s)	Visibility (10m)	Solar Radiation (MJ/m2)
1.299176	1.699290	1.932031
Rainfall(mm)	Snowfall (cm)	is_weekday
1.071697	1.113891	1.004875
is_holiday	is_winter	is_spring
1.024673	3.386889	1.711751
is_summer		
2.604600		

The results for VIF illustrate the multicollinearity of each independent variable with other independent variables. The variables Hour, Wind speed, Humidity, Visibility, Solar Radiation, Rainfall, Snowfall, is_weekday, is_holiday, is_winter, is_spring, and is_summer all have a range of values that are greater than 1 and less than 4, which shows that

each variable has a moderate correlation with the other independent variables. Additionally, the variable with the highest effect is Temperature at 5.1, indicating that it has a strong correlation with the other independent variables.

Results from Summary method:

Residuals:

Min	1Q	Median	3Q	Max
-1228.41	-276.09	-60.28	214.17	2257.95

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.766e+02	3.965e+01	17.066	< 2e-16	***
Hour	2.867e+01	7.477e-01	38.346	< 2e-16	***
`Temperature(°C)`	2.736e+01	8.803e-01	31.084	< 2e-16	***
`Humidity(%)`	-8.431e+00	3.732e-01	-22.590	< 2e-16	***
`Wind speed (m/s)`	1.594e+01	5.195e+00	3.069	0.00215	**
`Visibility (10m)`	6.771e-03	1.009e-02	0.671	0.50214	
`Solar Radiation (MJ/m2)`	-8.452e+01	7.547e+00	-11.199	< 2e-16	***
`Rainfall(mm)`	-6.363e+01	4.336e+00	-14.676	< 2e-16	***
`Snowfall (cm)`	3.289e+01	1.120e+01	2.936	0.00334	**
is_weekday	8.201e+01	1.043e+01	7.862	4.23e-15	***
is_holiday	-1.254e+02	2.228e+01	-5.630	1.86e-08	***
is_winter	-3.604e+02	1.990e+01	-18.111	< 2e-16	***
is_spring	-1.438e+02	1.415e+01	-10.166	< 2e-16	***
is_summer	-1.633e+02	1.732e+01	-9.425	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

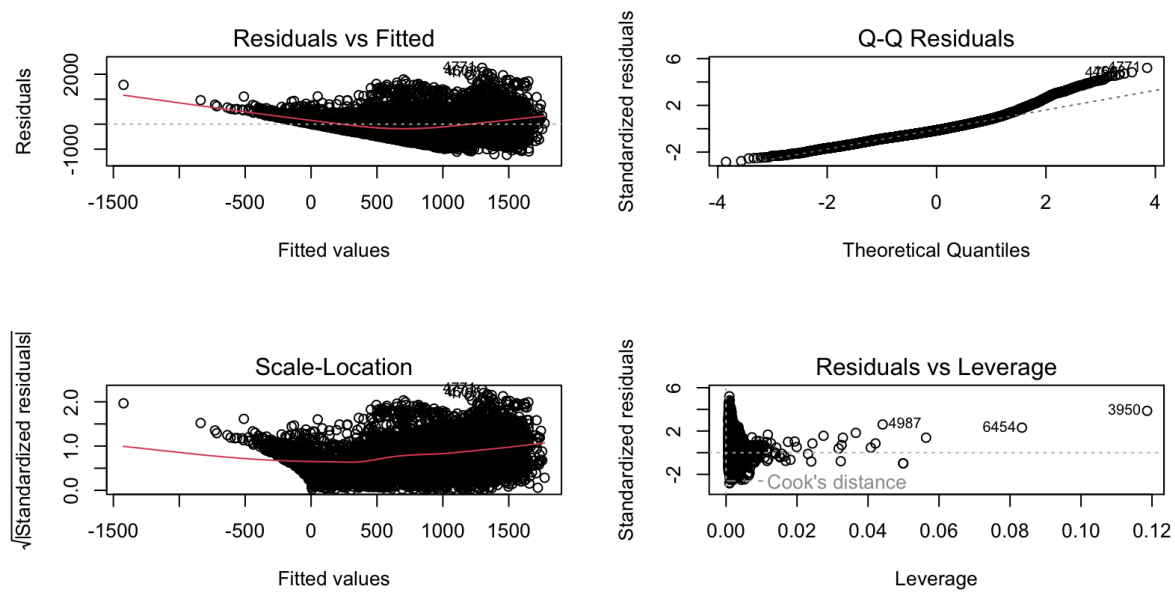
Residual standard error: 433.7 on 8451 degrees of freedom

Multiple R-squared: 0.5449, Adjusted R-squared: 0.5442

F-statistic: 778.3 on 13 and 8451 DF, p-value: < 2.2e-16

The regression model was statistically significant overall, with an adjusted R-squared score of 0.544, meaning the model explains approximately 54% of the variation in bike demand. All predictors' p-values were less than 0.05, except for Visibility (p=0.502). Therefore, the null hypothesis is rejected.

Plots:



The plot on the top left shows that as the number of fitted values increases, the residual increases; the spread of points with higher positive residuals or larger negative residuals increases. This indicates heteroscedasticity.

The Q-Q plot and scale-location showed mild deviations from normality and constant variance.

The residuals vs leverage plot shows no points with excessive influence.

Project Limitations:

1. The dataset is from 2018, so the predictions that are done do not reflect the current state of the bike demand market.
2. The dataset lacks bike allocation data, meaning recommendations can inform when to adjust fleet capacity, but cannot address where specifically bikes should be allocated across the city.

4. Recommendations & Conclusion:

This analysis confirms that the bike demand in Seoul is significantly influenced by season, weather conditions, holiday status, and weekends or weekdays. All five statistical tests, ANOVA (season), Welch's t-tests (snowfall, holiday, weekday/weekend), and multiple linear regression, showed statistically significant results, with the regression model explaining approximately 54% variance for bike demand.

Based on these findings, the company should switch from static, uniform fleet allocation and adopt a dynamic approach that adjusts bike availability based on expected conditions: higher fleet availability during warmer seasons and weekday peak hours, and less fleet availability during colder seasons and non-peak hours.

To conclude, these recommendations should be interpreted alongside limitations discussed above, mainly because the dataset age and lack of allocation data, which restricts the possibility of implementing these across the city.

5. Appendix: Screenshot of the code

```

1  install.packages(c("readr","lubridate","psych","dplyr","corplot","ggplot2",
2                        "car"))
3
4  library(readr)
5  library(lubridate)
6  library(psych)
7  library(dplyr)
8  library(corrplot)
9  library(ggplot2)
10 library(car)
11
12 #-----Data Loading & Exploration-----
13 df <- read_csv("SeoulBikeData.csv",
14               locale= locale(encoding = "Windows-1252"))
15
16 names(df)
17
18 head(df)
19 View(df)
20
21 str(df)
22
23 # Returns the number of null values
24 sum(is.na(df))
25 # Returns the number of duplicate rows
26 sum(duplicated(df))
27
28 # Returns the total number of values present in the 'Functioning Day' column
29 length_of_functioning_day <- length(df$`Functioning Day`)
30 # Returns the percentages of each unique value from 'Functioning Day'
31 percentage <- (table(df$`Functioning Day`)/length_of_functioning_day)*100
32 # Rounds to 2 decimal places
33 percentage <- round(percentage,2)
34 percentage
35
36 df_groupby_functioning_day <- df %>%
37   select(`Rented Bike Count`,`Functioning Day`) %>%
38   filter(`Functioning Day`=='No')
39 df_groupby_functioning_day
40
41 # -----Data Preprocessing -----
42 # Keeps rows where 'Functioning Day' column is equal to 'Yes'
43 df <- df %>%
44   filter(`Functioning Day`=='Yes')
45
46 # Displays the shape for the dataframe after filtering
47 dim(df)
48 # Drops the 'Dew point temperature(°C)' column and 'Functioning Day' column
49 # from the dataframe
50 df <- df %>%
51   select(!c(`Dew point temperature(°C)`,`Functioning Day`))
52
53 # Switches the column data format into date format
54 df <- df %>%
55   mutate(Date = as.Date(Date,format='%d/%m/%Y'))
56
57 # Returns a new column where days are numeric (Monday = 1, Sunday = 7)
58 df <- df %>%
59   mutate(dayofweek = wday(Date,label = TRUE))
60
61 # Creates a new column called 'is_weekday' that contain two values: 1 (True) and 0 (False)
62 df$is_weekday <- ifelse(df$dayofweek=='Fri'|df$dayofweek=='Mon'|df$dayofweek=='Tue'|df$dayofweek=='Wed'|df$dayofweek=='Thu',1,0)
63
64 # Creates a new column called 'is_holiday' for encoding and it contains two values:
65 # 1 (True) and 0 (False)
66 df$is_holiday <- ifelse(df$Holiday=='Holiday',1,0)

```

```

65
66 df$is_winter <- ifelse(df$Seasons=='Winter',1,0)
67 df$is_spring <- ifelse(df$Seasons=='Spring',1,0)
68 df$is_summer <- ifelse(df$Seasons=='Summer',1,0)
69
70 df$is_snowfall <-ifelse(df$`Snowfall (cm)`==0,"No snowfall","Snowfall")
71
72 # Converts the datatype of 'Rented Bike Count' column into numeric type
73 df <-df %>%
74   mutate(`Rented Bike Count` = as.numeric(`Rented Bike Count`))
75 # Drops the Holiday, and dayofweek columns
76 df <- df %>%
77   select(!c(Holiday,dayofweek))
78
79 df_numerical_columns <- df %>%
80   select(!c(Seasons,is_snowfall,Date))
81 # Returns a Statistical summary
82 describe(df_numerical_columns)
83
84 df_columns_for_corr <- cor(df_numerical_columns)
85
86 # Returns a dataframe with Seasons grouped and
87 # aggregated by the mean bike demand per season
88 df_groupby_season <- df %>%
89   group_by(Seasons) %>%
90   summarise(avgSeasonDemand=round(sum(`Rented Bike Count`)/length(Date)))
91
92 # Returns the overall average bike demand
93 avg_Demand<- df %>%
94   summarise(avgDemand=round(sum(`Rented Bike Count`)/length(Date)))
95
96 df_grouped_by_season_and_hour <- df %>%
97   summarise(avgHourDemand=round(mean(`Rented Bike Count`))
98     ,.by=c(Seasons,Hour))
99
100 df_grouped_by_season_and_temp <- df %>%
101   summarise(avgDemand=round(mean(`Rented Bike Count`))
102     ,.by=c(Seasons,`Temperature(°C)`))
103
104 df_inWinter <-df %>%
105   filter(Seasons=='Winter')
106

```

```

107 # ----- EDA -----
108
109 # Displays a histogram based on 'Rented Bike Count' column
110 hist(df$`Rented Bike Count`,main=paste("Histogram of",
111    "Rented Bikes"),xlab='Number of rented bikes',col="lightblue")
112
113 # Displays a correlation matrix
114 corplot(df_columns_for_corr)
115 #----- Data Viz for business stakeholders -----
116 avg_Demand=unlist(avg_Demand)
117
118 # Displays a bar chart for the Average daily bike demand per Season
119 ggplot(df_groupby_season,aes(x=Seasons,y= avgSeasonDemand
120    ,fill = Seasons)) +
121   geom_col() +
122   labs(x = "Seasons", y = "Average demand") +
123   geom_hline(yintercept = avg_Demand,linetype = "dashed") +
124   geom_text(aes(label = avgSeasonDemand),vjust=-0.4, colour = "black")+
125   theme_minimal() +
126   ggtitle("Average Daily Bike Demand across Multiple Seasons")
127
128 # Displays a linechart where x is the hour and y is the average demand per hour, colored by a line representing the Seasons
129 ggplot(df_grouped_by_season_and_hour,aes(x=Hour,y=avgHourDemand
130    ,color=Seasons)) +
131   geom_line() +
132   expand_limits(y=0) +
133   theme_minimal() +
134   labs(x = "Hour of the day", y = "Average demand") +
135   ggtitle("Average Hourly Bike Demand across Multiple Seasons")
136
137 # Displays a scatterplot where x is temperature and y is the average bike demand
138 ggplot(df_grouped_by_season_and_temp,
139    aes(x=`Temperature('C)`',y=avgDemand,color=Seasons)) +
140   geom_point() +
141   theme_minimal() +
142   labs(x = "Temperature", y = "Average demand") +
143   ggtitle("Temperature vs Average Demand (across seasons)")
144
145
146
147 # Displays a boxplot that consists of 2 plots
148 ggplot(df_inWinter,aes(x=is_snowfall,y=`Rented Bike Count`,
149    color=is_snowfall)) +
150   geom_boxplot(outlier.shape = NA) +
151   scale_y_log10() +
152   #coord_flip() +
153   theme_minimal() +
154   labs(x = "Snowfall or No Snowfall", y = "Average demand") +
155   ggtitle("Comparing the demand when it is snowing vs when there is no snow")
156
157 #-----Hypothesis Testing -----
158
159 # Test 1: Anova
160 df_aov <- aov(df$`Rented Bike Count`~ df$Seasons)
161 summary(df_aov)
162
163 # Tukey significant difference test
164 TukeyHSD(df_aov)
165

```

```

166 # Test 2: T-test for 'is_snowfall'
167 t.test(`Rented Bike Count`~is_snowfall,data=df_inWinter)
168 # Test 3: T-test for 'is_holiday'
169 t.test(`Rented Bike Count`~is_holiday,data=df)
170
171 # Test 4: T-test for 'is_weekday'
172 t.test(`Rented Bike Count`~is_weekday,data=df)
173
174 #----- Modeling -----
175
176 df_linear_regression <- df %>%
177   select(!c(Date,Seasons,is_snowfall))
178 # Multivariable linear Regression Model
179 Linear_regression_model <- lm(`Rented Bike Count`~.,df_linear_regression)
180
181 # Returns the calculate Variance Inflation Factor
182 vif(Linear_regression_model)
183
184 summary(Linear_regression_model)
185
186 par(mfrow = c(2,2))
187
188 plot(Linear_regression_model)
189
190 # Create a dataframe with new values
191 prediction_data <- data.frame(Hour=9,`Temperature(°C)`= 6.0,`Humidity(%)`=34,
192                               `Wind speed (m/s)`=3,`Visibility (10m)`=400,
193                               `Solar Radiation (MJ/m2)`=1.12,`Rainfall(mm)`=0,
194                               `Snowfall (cm)`=0,is_weekday=1,is_holiday=0,
195                               is_winter=1,is_spring=0,is_summer=0,
196                               check.names=FALSE)
197 # Gets the predicted value
198 Linear_Regression_prediction <- predict(Linear_regression_model
199                                       ,prediction_data)
200 # Displays the predicted value
201 Linear_Regression_prediction

```

Displayed output for prediction:

```

Browse[1]> Linear_Regression_prediction
1
489.7248

```

6. References:

1. Siavoshi, M. (2024). 'Step-by-Step Guide to Linear Regression in R', *Statology*, Available at:
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4. Kassambara, A. (2020) *ggplot2: Data Visualization in R*. Datanovia.
<https://www.datanovia.com/learn/data-visualization/ggplot2/>